

Unit 4.2 The mean value theorem p 284

Note Title

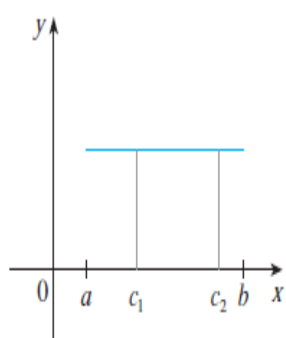
2014/04/06

Rolle's Theorem p284 (No proof)

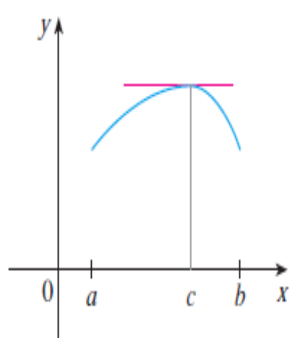
f is a function so that:

- ① f continuous on $[a, b]$
- ② f differentiable on (a, b)
- ③ $f(a) = f(b)$

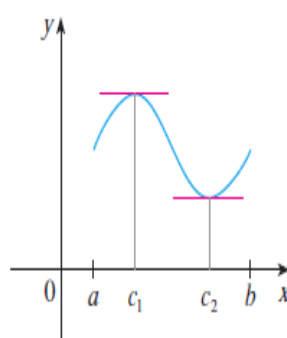
then there exists a number c so that $f'(c) = 0$



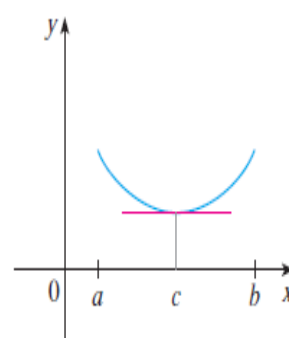
(a)



(b)



(c)



(d)

Mean value theorem p285 (No proof) (MVT)

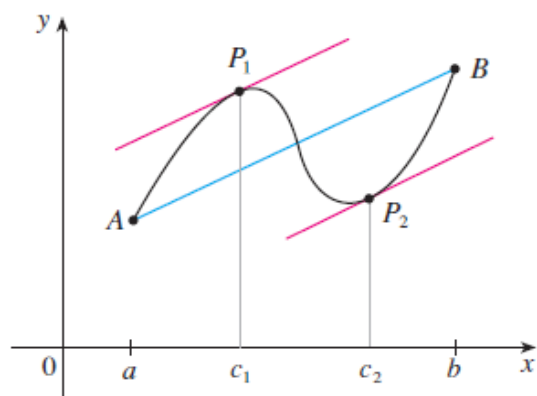
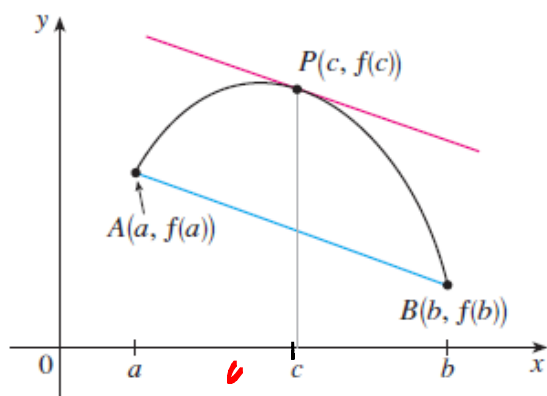
f is a function so that:

- ① f continuous on $[a, b]$
- ② f differentiable on (a, b)

then there exists a number $c \in (a, b)$

so that $f'(c) = \frac{f(b) - f(a)}{b - a}$ or

$$f(b) - f(a) = f'(c)(b - a)$$



Examples

Verify the two conditions required by the MVT and find a suitable c guaranteed to exist if $P(x) = x^2 + 5x$ on $[0, 2]$

1. P is continuous on $[0, 2]$ since

P is a polynomial

2. P is differentiable on (a, b) since

$$P'(x) = 2x + 5 \quad \text{which means}$$

$P'(x)$ exists on $(0, 2)$

MVT says there exists a number

$c \in (0, 2)$ so that

$$P'(c) = \frac{P(2) - P(0)}{2 - 0} = \frac{14 - 0}{2 - 0} = 7$$

$$\text{Also } P'(c) = 2c + 5 = 7$$

$$\Rightarrow \underline{c = 1} \quad \triangleright$$